Chapter 6

CIRCULAR MOTION AND OTHER APPLICATIONS OF NEWTON'S LAWS

Chuyển động tròn và những ứng dụng khác của các định luật Newton

OUTLINES

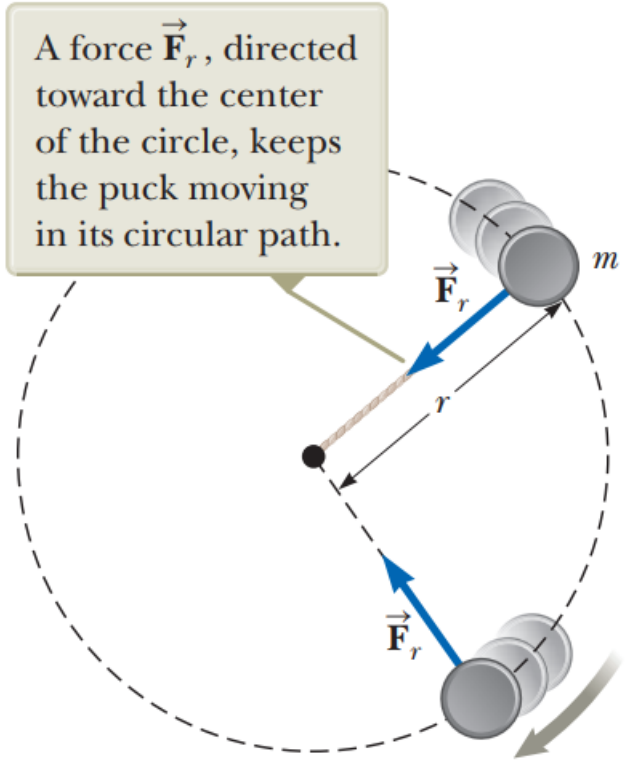
- ❑Uniform Circular Motion
- ❑Nonuniform Circular Motion
- ❑ Motion in Accelerated Frame
- ❑Fictitious force
- ❑ Motion in the Presence of Resistive Forces

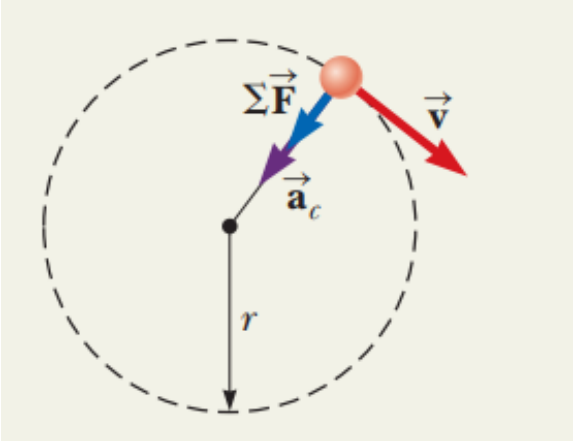
Particle in Uniform Circular Motion Model

Consider a particle moving in a circle of radius r with a constant speed v in the horizontal plane:

centripetal acceleration

$$a_c = \frac{v^2}{r}$$





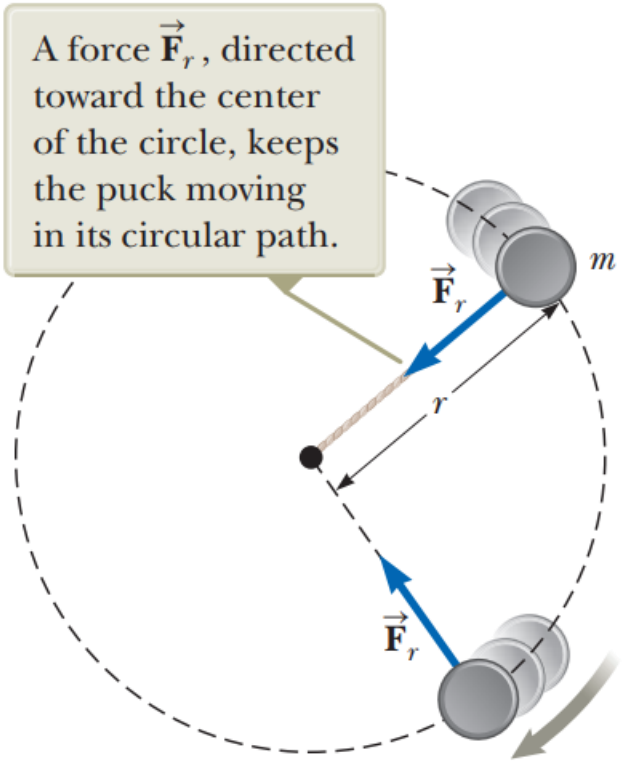
- The string exerts a radial force on the puck to keep it move in a circular path.
- Newton's second law along the radial direction:

$$\sum F_r = ma_c = m\frac{v^2}{r}$$

F_r causes a centripetal acceleration a_c and causes a change in the direction of the velocity vector

Extending the Particle in Uniform Circular Motion Model

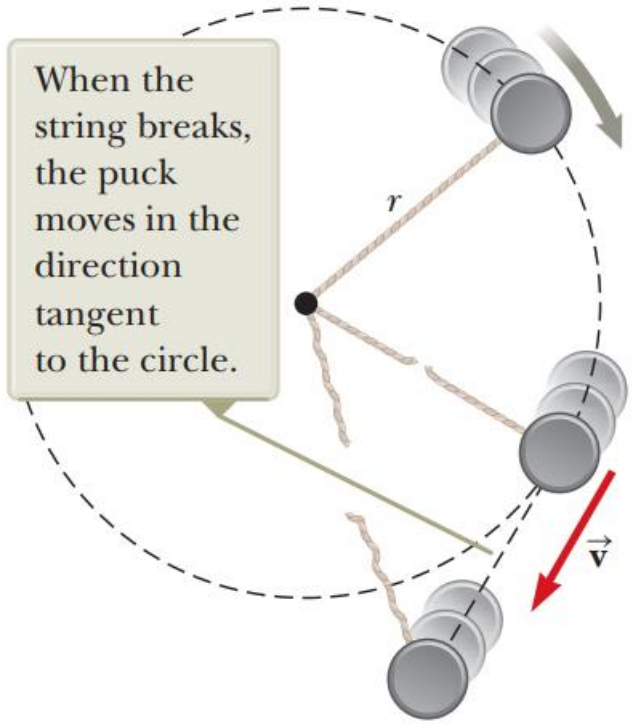
If F_r vanishes => the object will move along a straight-line path tangent to the circle, instead of moving in a circular path



It moves in a circular path

The string breaks

Without F_r

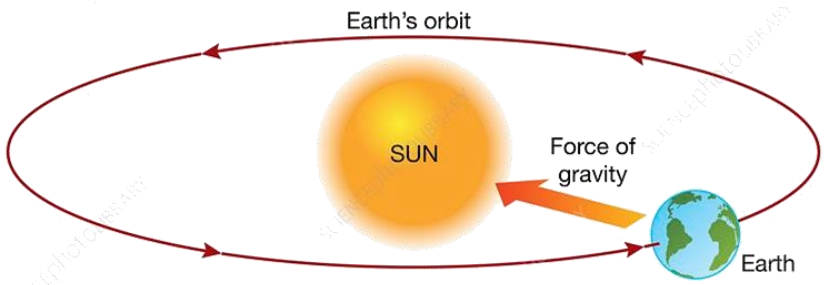


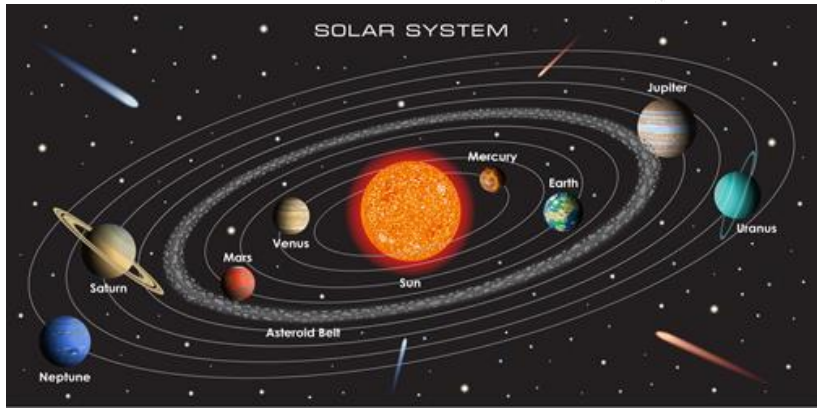
It moves in a straight-line path tangent to the circle

EXAMPLES OF CENTRIPETAL FORCE

Centripetal force takes place along a circular motion. Its direction is towards the centre of the curvature

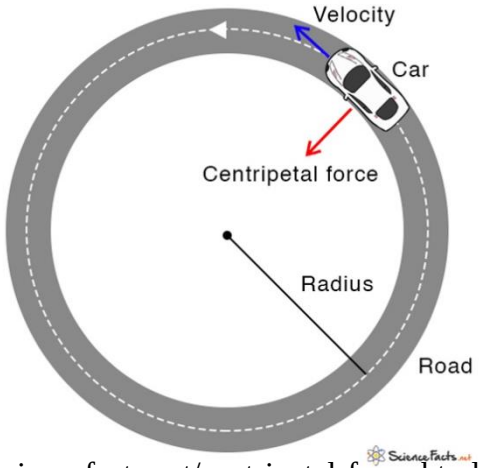
Centripetal force as a gravity force



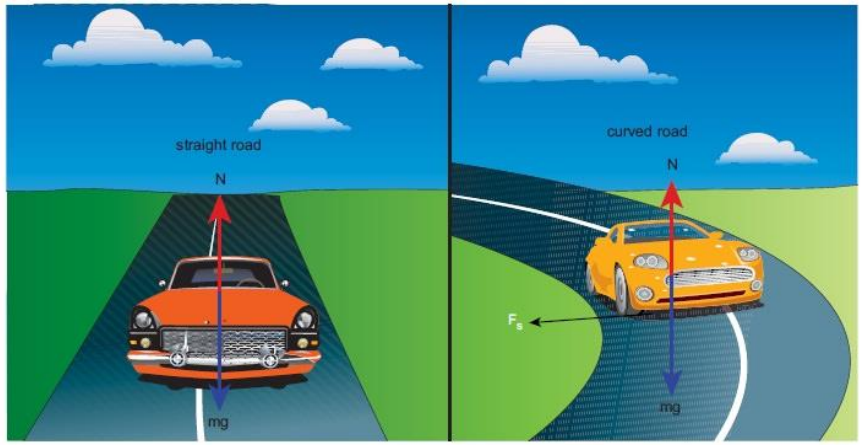


https://www.yaclass.in/p/science-cbse/class-9/gravitation-8367/universal-law-of-gravitation-8646/re-8b6c7782-c9c1-4abd-96b7-0ada9fb247b7

Centripetal force as friction force

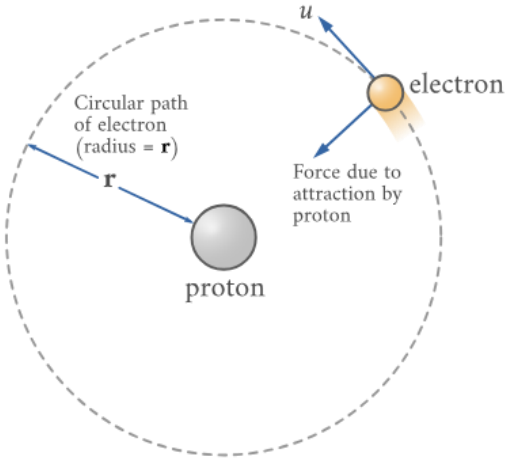


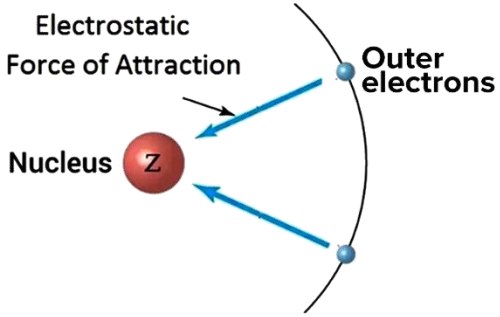
https://www.sciencefacts.net/centripetal-force.html



https://chem.libretexts.org/Bookshelves/General_Chemistry/Map%3A_Chemistry_(Zumdahl_and_Decoste)/07%3A_Atomic_Structure_and_Periodicity/12.04_The_Bohr_Model

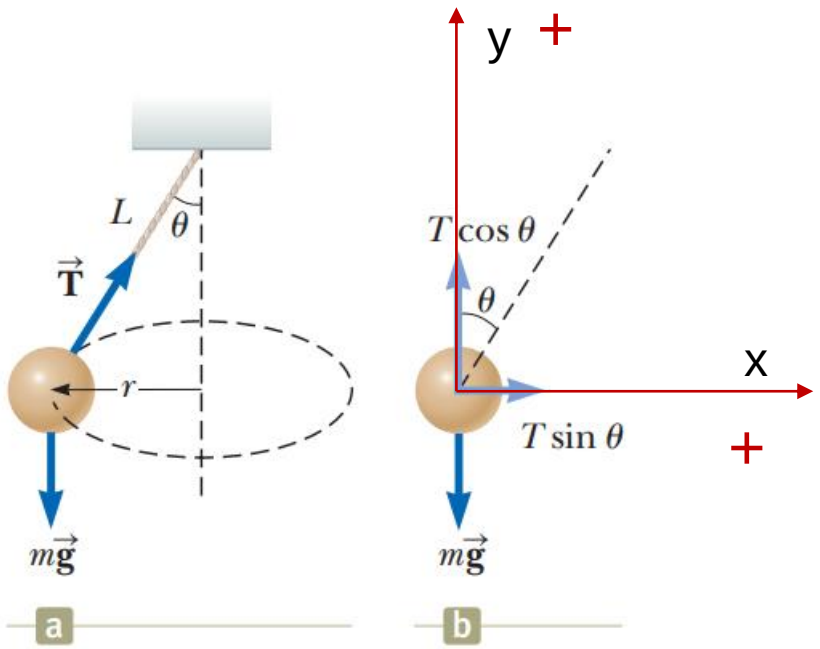
Centripetal force as electric force





Example 6.1 The Conical Pendulum AM

A small ball of mass m is suspended from a string of length L. The ball revolves with constant speed v in a horizontal circle of radius r as shown in Figure 6.3. (Because the string sweeps out the surface of a cone, the system is known as a *conical pendulum*.) Find an expression for v in terms of the geometry in Figure 6.3.



Solutions:

Applying Newton's second law:

+ in x direction:

$$\sum \vec{F}_x = m\vec{a}_c$$

x → + : $T\sin\theta = ma_c = m\frac{v^2}{r}$ (1)

+ in y direction: $\sum \vec{F}_y = m\vec{a}_y = 0$

y ↑ + : $T\cos\theta - mg = 0 \Rightarrow T = \frac{mg}{\cos\theta}$ (2)

Solutions:

Substituting eq (2) into eq(1):

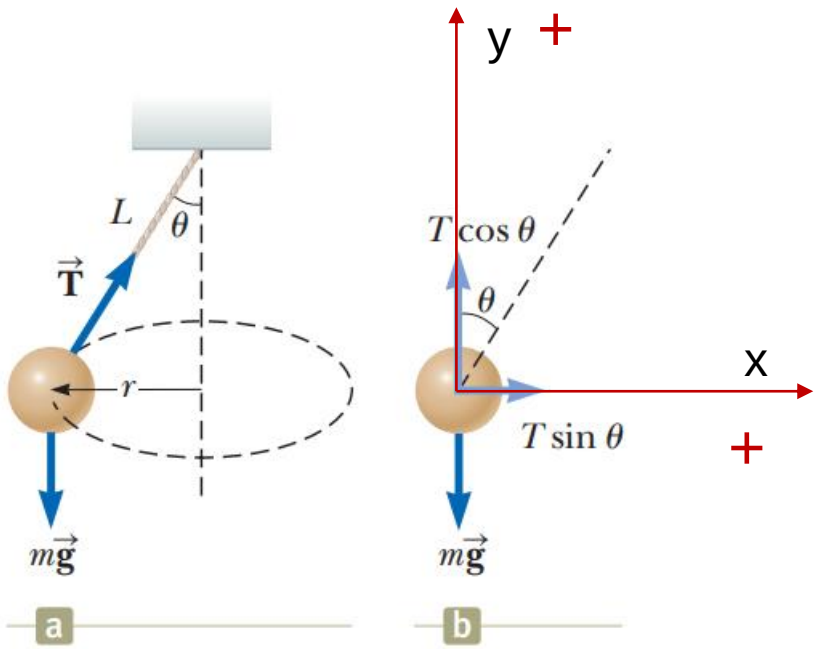
$$\frac{mgsin\theta}{cos\theta} = m\frac{v^2}{r}$$

$$\Rightarrow v = \sqrt{grtan\theta} \quad (3)$$

In addition: $\sin\theta = \frac{r}{L}$ => r = sinθ.L

Substituting r into eq (3):

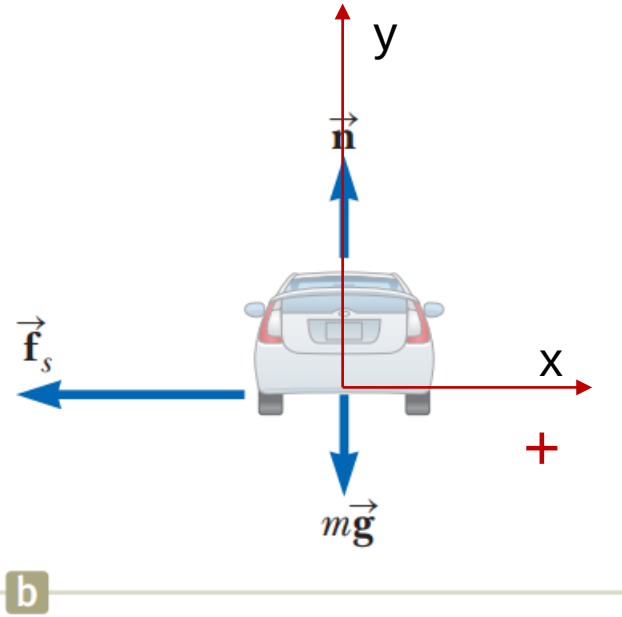
$$v = \sqrt{gLsin\theta tan\theta}$$



Example 6.3 What Is the Maximum Speed of the Car? AM

A 1 500-kg car moving on a flat, horizontal road negotiates a curve as shown in Figure 6.4a. If the radius of the curve is 35.0 m and the coefficient of static friction between the tires and dry pavement is 0.523, find the maximum speed the car can have and still make the turn successfully.

r=35m, m = 1500kg, μ_s=0.523



Solutions:

Applying Newton's second law:

- In x direction:

$$\sum \vec{F}_x = m\vec{a}_c$$

x →
+

$$-f_s = -ma_c = -m\frac{v^2}{r}$$

$$\Rightarrow f_s = m\frac{v^2}{r} \quad (1)$$

$$f_s = m\frac{v^2}{r} \Rightarrow f_{s,max} = m\frac{{v_{max}}^2}{r} \quad (2)$$

Applying Newton's second law:

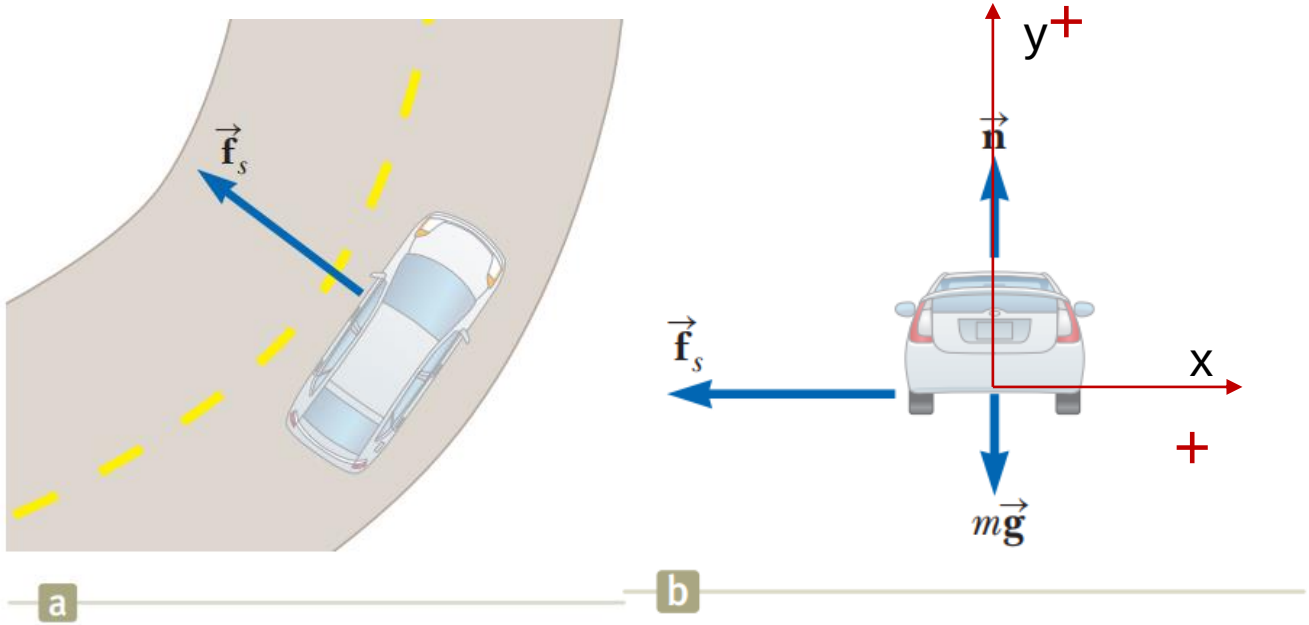
- In y direction:

$$\sum \vec{F}_y = m\vec{a}_y = 0$$

$$\Leftrightarrow \vec{n} + \vec{F_g} = 0$$

y+

$$n - mg = 0 \Rightarrow n = mg \quad (3)$$



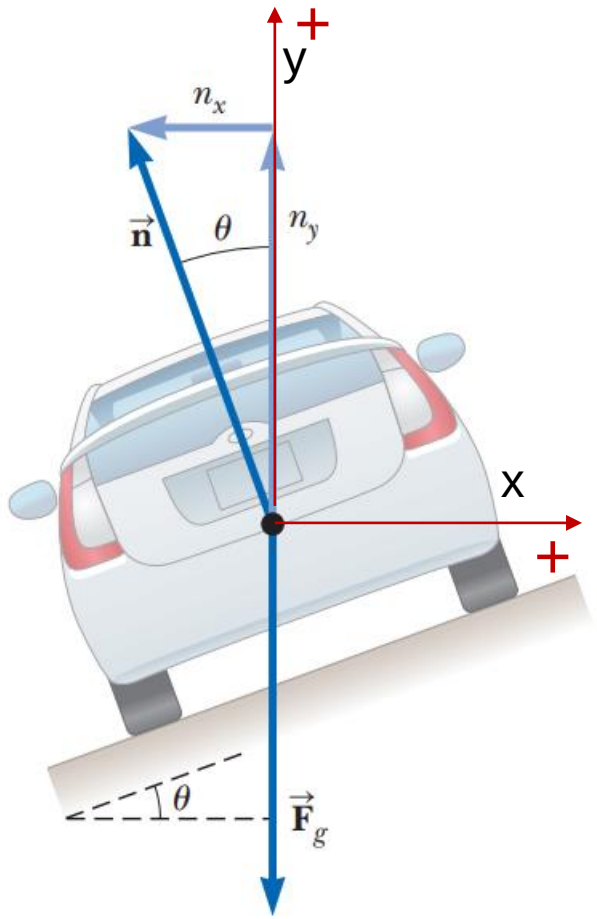
In addition: $f_{s,max} = \mu_s n = \mu_s\, mg$

The maximum speed of the car:

From eq(2): $v_{max} = \sqrt{\frac{f_{s,max} r}{m}} = \sqrt{\frac{\mu_s m g r}{m}} = \sqrt{g r \mu_s} = \sqrt{9.8\text{x}35\text{x}0.523} = 13\ m/s$

Example 6.4 The Banked Roadway AM

A civil engineer wishes to redesign the curved roadway in Example 6.3 in such a way that a car will not have to rely on friction to round the curve without skidding. In other words, a car moving at the designated speed can negotiate the curve even when the road is covered with ice. Such a road is usually *banked*, which means that the roadway is tilted toward the inside of the curve as seen in the opening photograph for this chapter. Suppose the designated speed for the road is to be 13.4 m/s (30.0 mi/h) and the radius of the curve is 35.0 m. At what angle should the curve be banked?



Solutions:

Applying Newton's second law:

- In x direction:

$$\sum \vec{F}_x = m\vec{a}_c \xrightarrow[+]{x} -n_x = -ma_c$$

$$\Leftrightarrow n_x = m\frac{v^2}{r} \Leftrightarrow n\sin\theta = m\frac{v^2}{r} \quad (1)$$

Applying Newton's second law:

- In y direction:

$$\sum \vec{F}_y = m\vec{a}_y = 0 \quad \uparrow y^+ \quad n_y - F_g = 0 \Leftrightarrow n\cos\theta = mg \quad (2)$$

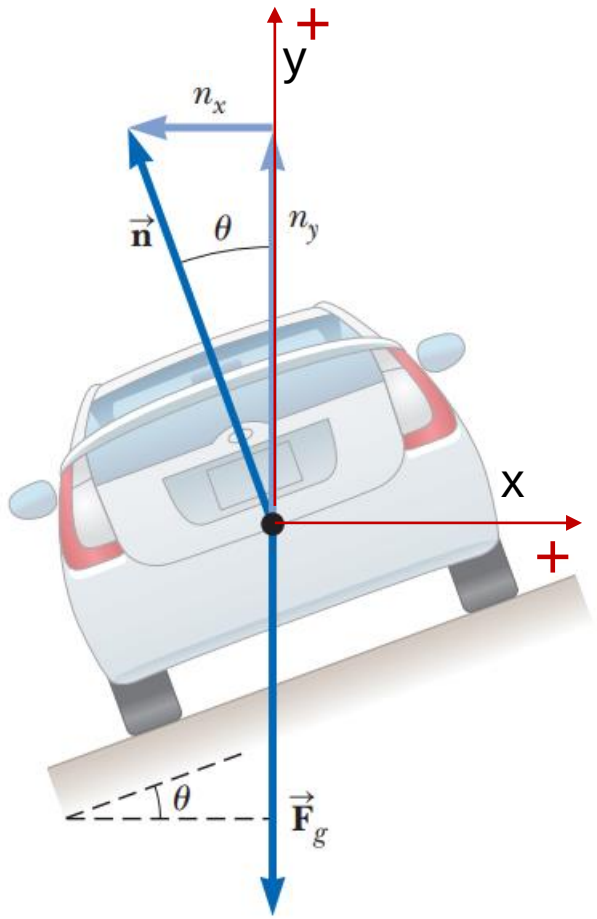
$n\sin\theta = m\frac{v^2}{r}$ (1)

$n\cos\theta = mg$ (2)

Dividing eq(1) by eq(2):

$$tan\theta = \frac{v^2}{gr} = \frac{13.4^2}{9.8x35}$$

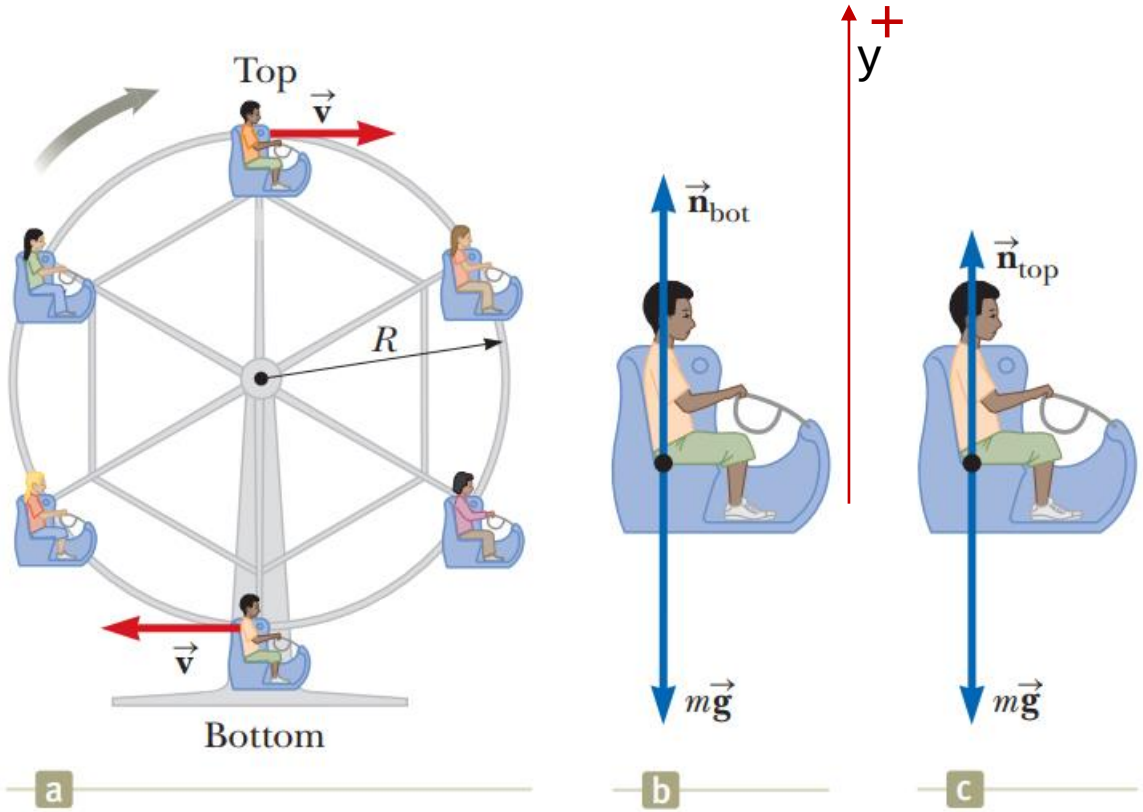
=> $\theta = 27.6°$



Example 6.5 Riding the Ferris Wheel AM

A child of mass m rides on a Ferris wheel as shown in Figure 6.6a. The child moves in a vertical circle of radius 10.0 m at a constant speed of 3.00 m/s.

(A) Determine the force exerted by the seat on the child at the bottom of the ride. Express your answer in terms of the weight of the child, mg.



Solutions:

Applying Newton's second law:

- In y direction:

$$\sum \vec{F}_y = m\vec{a}_y$$

y+

$$n - mg = ma_c$$

$$\Leftrightarrow n = mg + m\frac{v^2}{2r} = mg\left(1 + \frac{v^2}{2gr}\right) = 1.09\ mg$$

(B) Determine the force exerted by the seat on the child at the top of the ride.

Solutions:

Applying Newton's second law:

- In y direction:

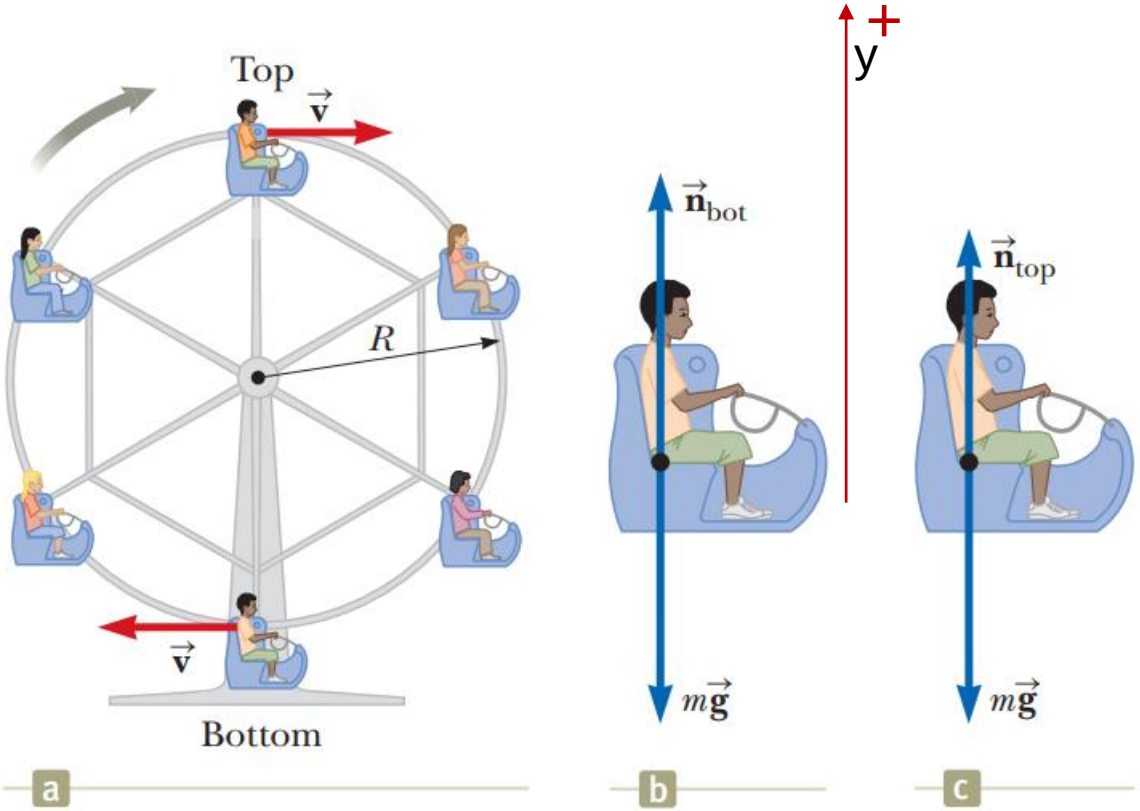
$$\sum \vec{F}_y = m\vec{a}_y$$

y+

$$n - mg = -ma_c$$

$$\Leftrightarrow n = mg - m\frac{v^2}{2r}$$

$$= mg\left(1 - \frac{v^2}{2gr}\right) = 0.9\ mg$$



Nonuniform Circular Motion

If particle moves with a varying speed:

- The acceleration consists of radial component of acceleration and a tangential component:

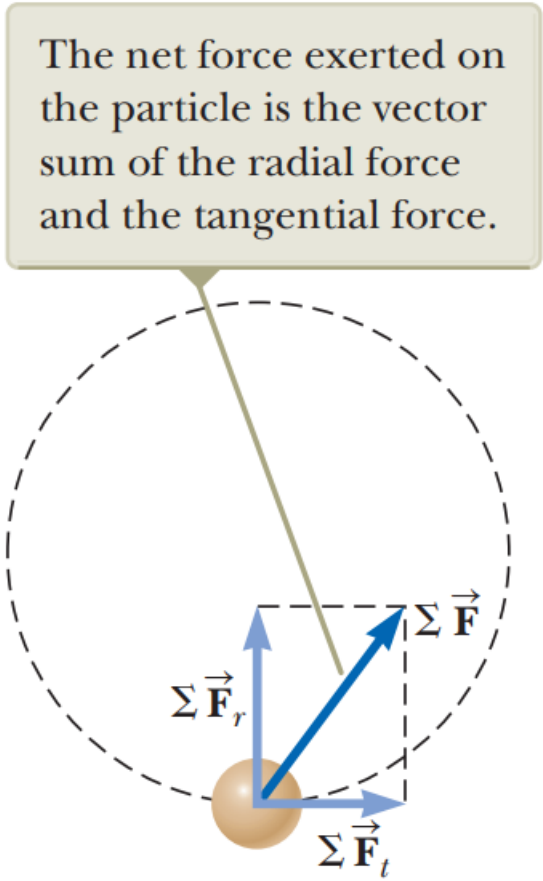
$$\vec{a} = \overrightarrow{a_t} + \overrightarrow{a_r}$$

- The total force applied to the particle:

$$\sum \vec{\mathbf{F}} = \sum \vec{\mathbf{F}}_r + \sum \vec{\mathbf{F}}_t$$

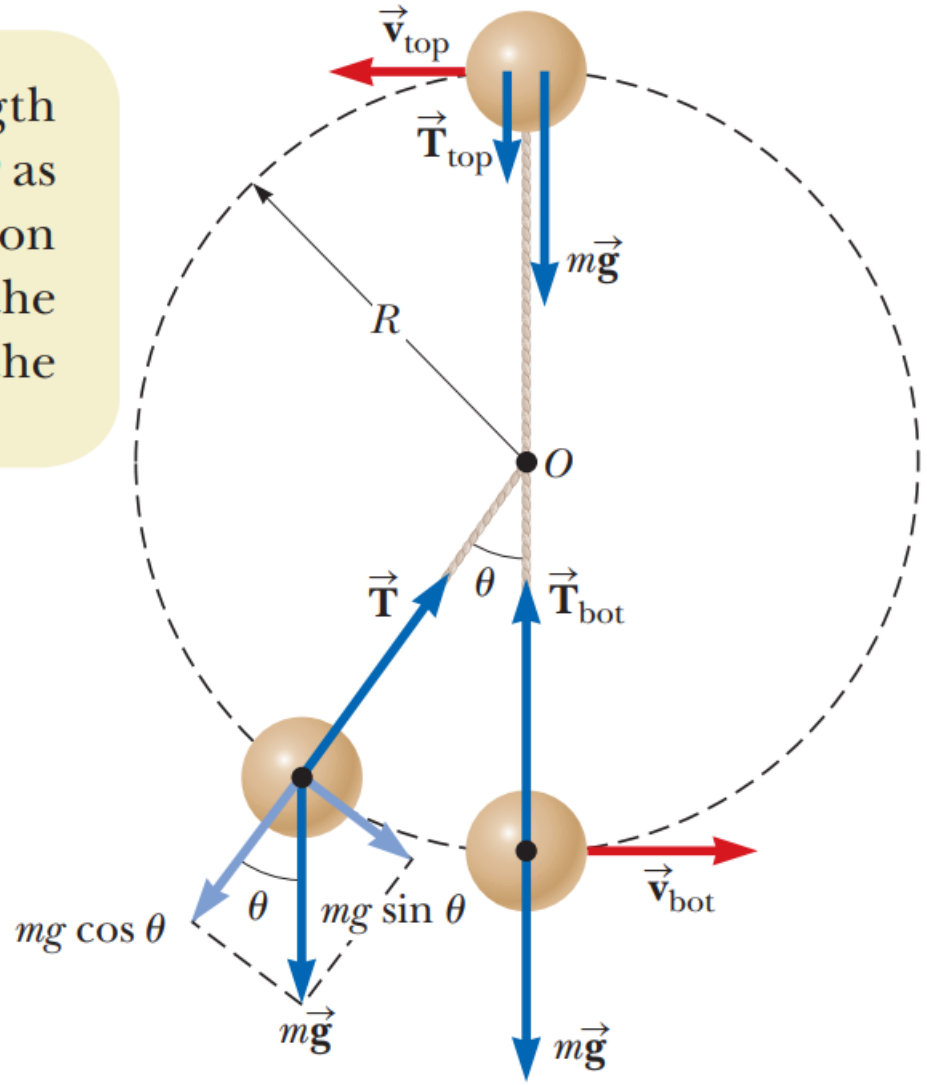
$\overrightarrow{F_r}$: radial force: directed toward the center of the circle

$\overrightarrow{F_t}$: tangential force: tangent to the circle



Example 6.6 Keep Your Eye on the Ball AM

A small sphere of mass m is attached to the end of a cord of length R and set into motion in a *vertical* circle about a fixed point O as illustrated in Figure 6.9. Determine the tangential acceleration of the sphere and the tension in the cord at any instant when the speed of the sphere is v and the cord makes an angle θ with the vertical.



Solutions:

Applying Newton's second law:

- In tangential direction:

$$\sum \vec{F}_t = m\vec{a}_t$$

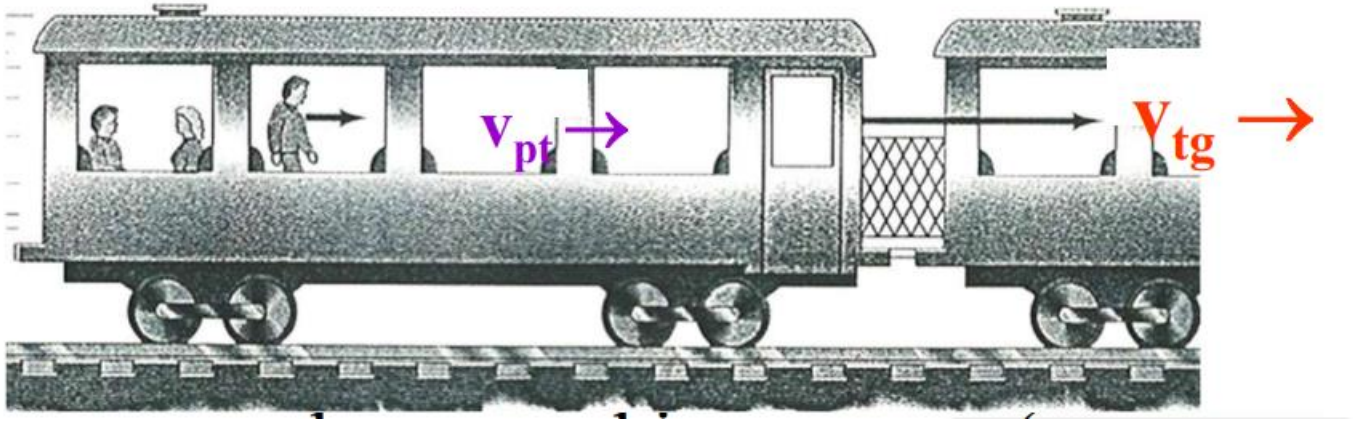
$$+ \quad mgsin\theta = ma_t \Rightarrow a_t = gsin\theta$$

- In radial direction: $\sum \vec{F}_r = m\vec{a}_r$

$$+ \quad T - mgcos\theta = ma_r \Rightarrow T = m\frac{v^2}{R} + mgcos\theta$$

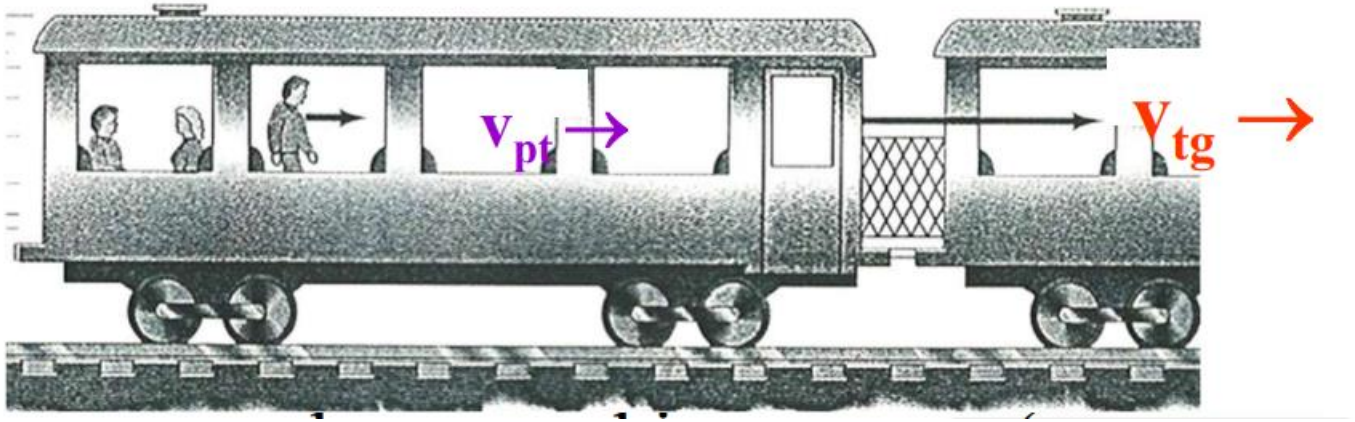
Motion in Accelerated Frame

- Newton's Laws technically apply only in inertial (non-accelerated) reference frames.
- We will analyze the problems in a noninertial frame of reference (accelerated reference):
- Consider a train moving with respect to the ground at **constant velocity** v_{tg} to the right. A person is walking in the train at a **constant velocity** v_{pt} with respect to the train:



If the train is **accelerated (v_{tg} # constant)**: the person will **undergo an acceleration** in the opposite direction

Motion in Accelerated Frame

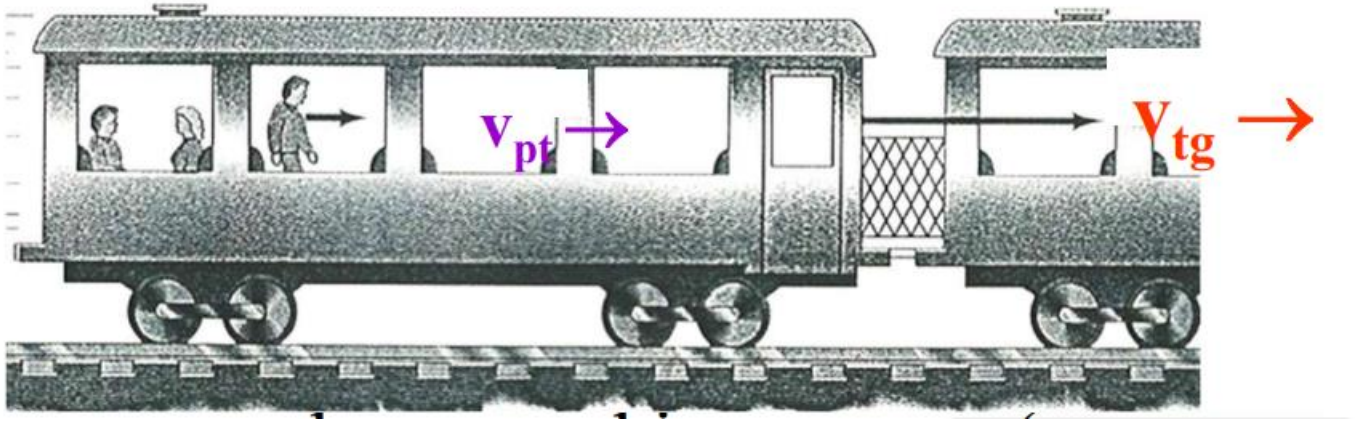


- If the train is **accelerated**: the person will **undergo an acceleration** in the opposite direction.
- However, according to Newton's first law: in the absence of unbalanced forces, the acceleration should be zero

WHY NEWTON'S LAWs ARE VIOLATED IN THIS CASE?

Answer: because the reference frame is non-inertial frame (which is a accelerated train)

Motion in Accelerated Frame



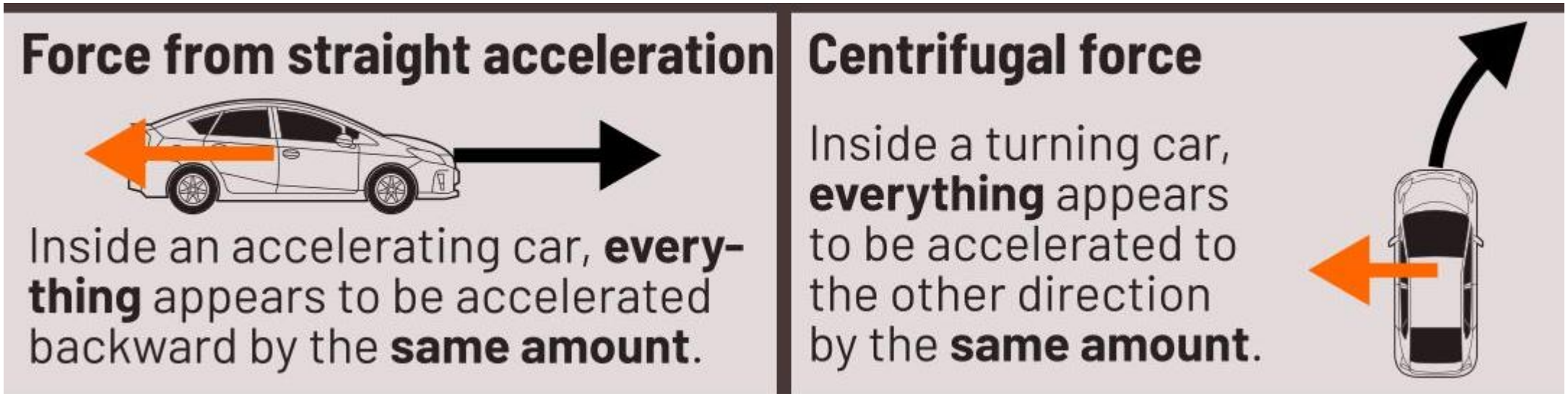
Q: Which force cause the person to accelerate?

fictitious force (lực quán tính).

This force is not a real force and only due to observations made in **non-inertial** frame (accelerated frame)

Fictitious force

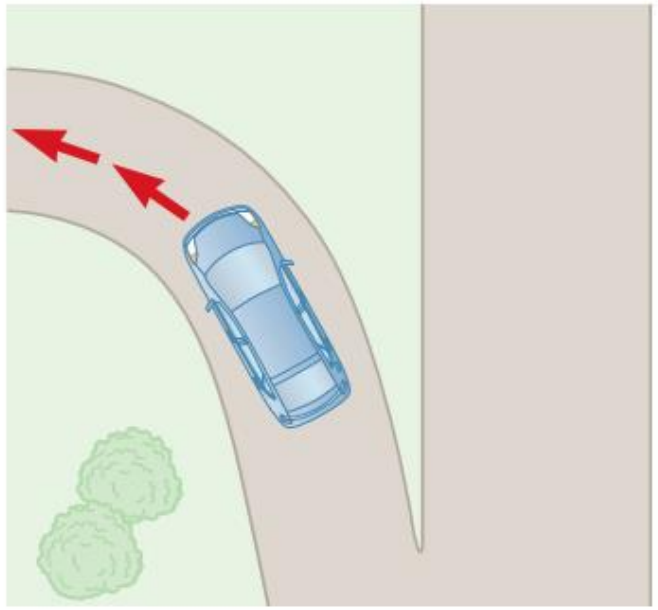
- A fictitious force appears to act on every object when the frame of reference is non-inertial (accelerating).
- A “fictitious” force does NOT mean the force does NOT exist
- A fictitious” force mean it does NOT arise from an interaction between objects



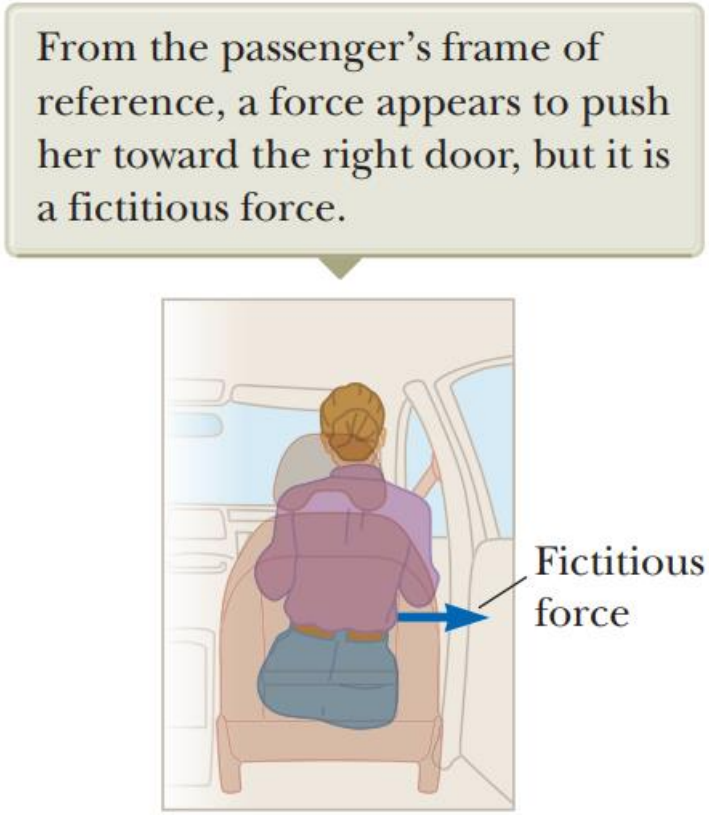
Fictitious force

- fictitious force is due to the change in the direction of the velocity vector.

The car changes the direction



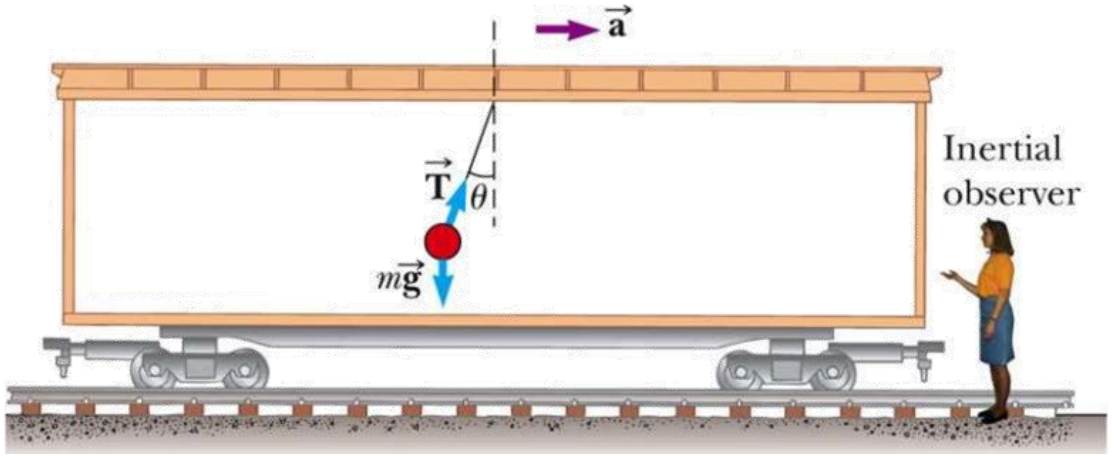
a



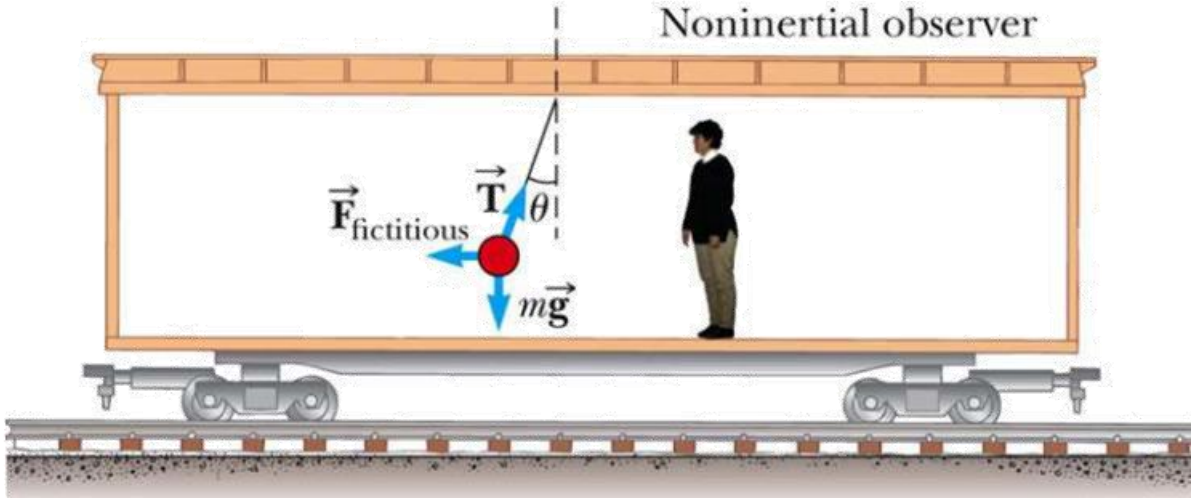
b

Fictitious force

- fictitious forces act in the direction opposite that of the acceleration of the noninertial frame



For inertial observer:
the object is moving with acceleration $\vec{a}$.
The object does NOT hang straight down. WHY?
=> the tension provides a horizontal component to give the object an acceleration.



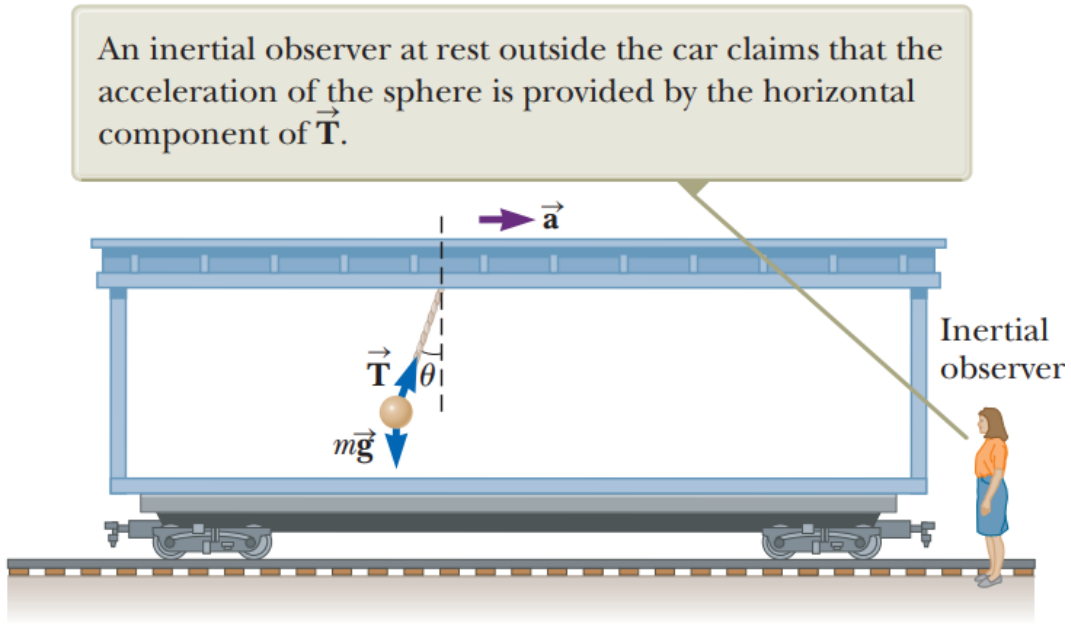
For noninertial observer: the object is at rest.
However, the hanging object still does NOT hang straight down. WHY???
=> Fictitious force pulling this object out to one side.

https://www.ux1.eiu.edu/~addavis/1350/06CirMtn/AccFrames.html

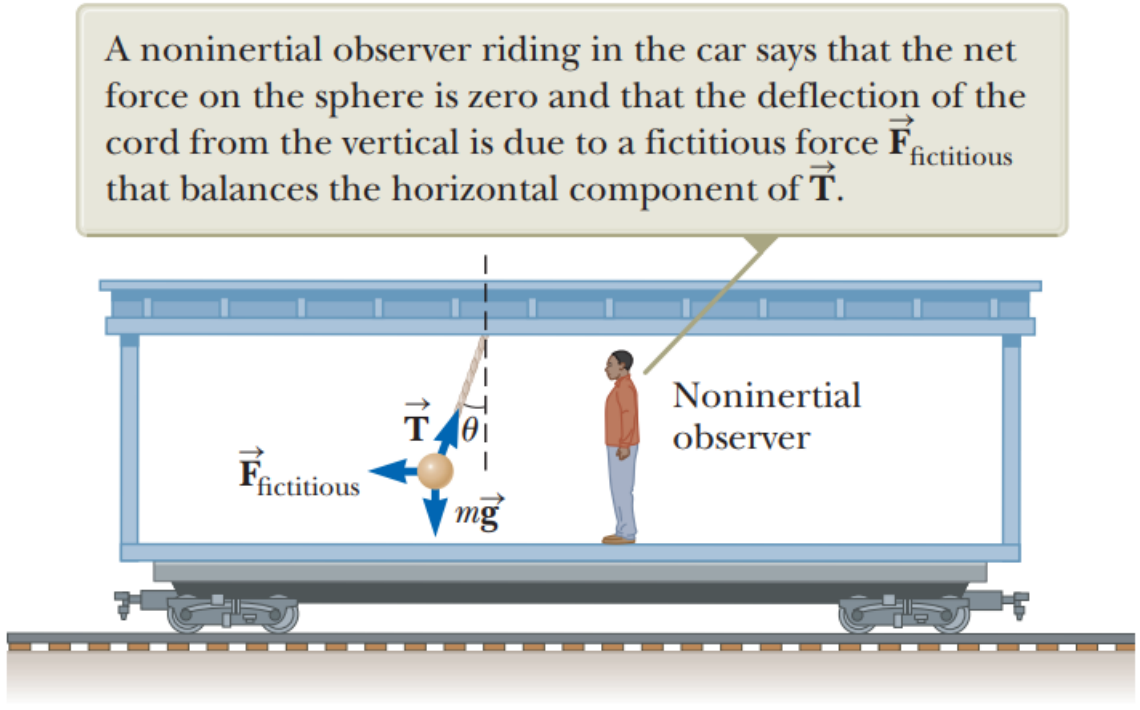
Example 6.7 Fictitious Forces in Linear Motion AM

A small sphere of mass m hangs by a cord from the ceiling of a boxcar that is accelerating to the right as shown in Figure 6.12. Both the inertial observer on the ground in Figure 6.12a and the noninertial observer on the train in Figure 6.12b agree that the cord makes an angle θ with respect to the vertical. The noninertial observer claims that a force, which we know to be fictitious, causes the observed deviation of the cord from the vertical. How is the magnitude of this force related to the boxcar's acceleration measured by the inertial observer in Figure 6.12a?

6.7 continued



a



b

How is the magnitude of fictitious force related to the boxcar's acceleration measured by the inertial observer in Figure 6.12a?

Solutions:

For the inertial observer: the sphere is accelerating

Applying Newton's second law:

- In x direction:

$$\sum \vec{F}_x = m\vec{a}$$

x

$$T sin\theta = ma \ (1)$$

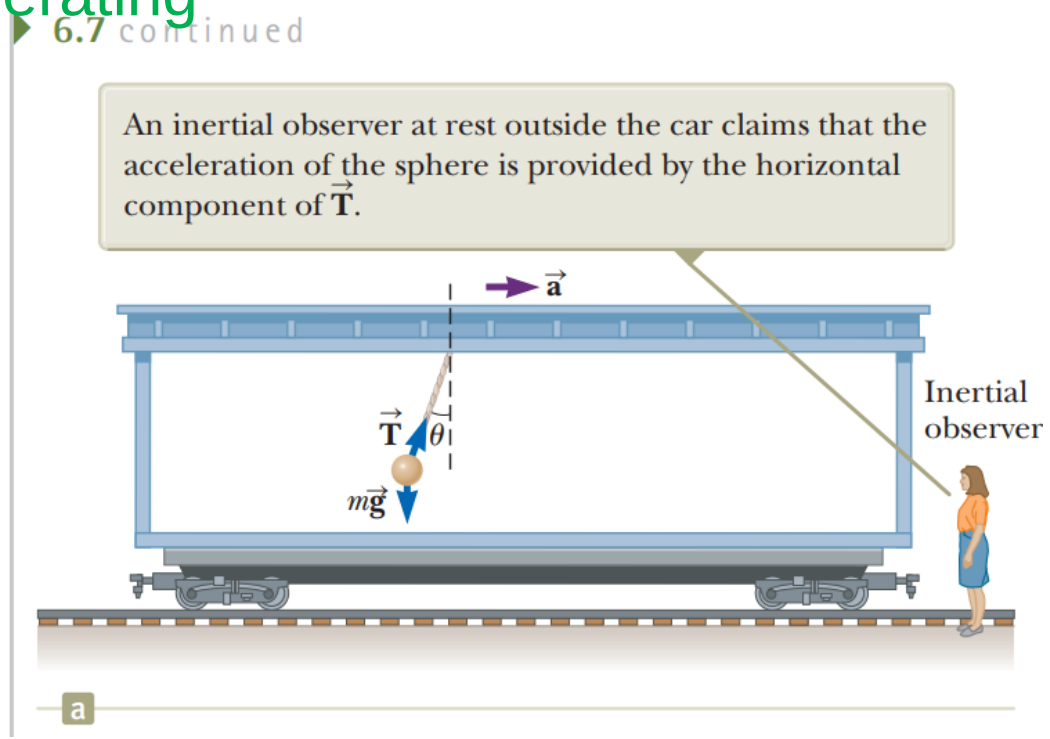
+

- In y direction:

$$\sum \vec{F}_y = m\vec{a}_y = 0$$

y+

$$T cos\theta - mg = 0 \Rightarrow T = \frac{mg}{cos\theta} \ (2)$$



From eq(1) and (2):

=> $g tan\theta = a(*)$

Solutions:

For the noninertial observer: the sphere is at rest

Applying Newton's second law:

- In x direction:

$$\sum \vec{F}_x = m\vec{a} = 0$$

x

+

$$T sin\theta - F_{fictitious} = 0$$

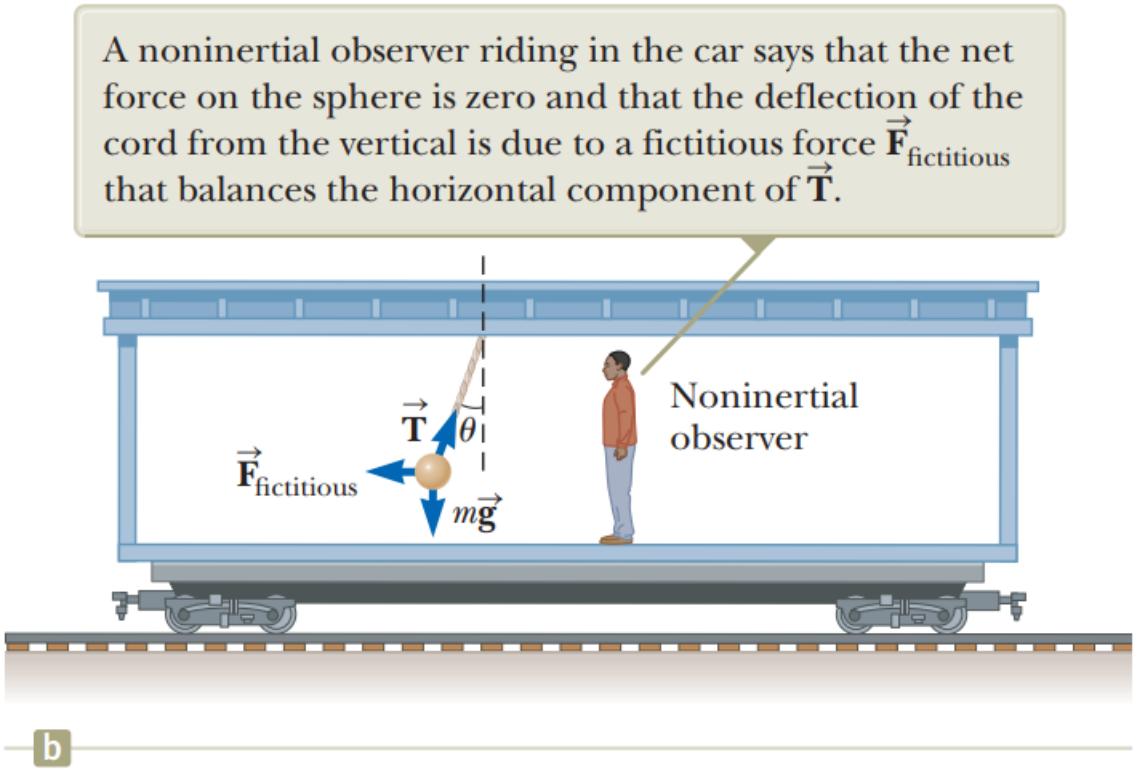
$$=> F_{fictitious} = T sin\theta \ (3)$$

- In y direction:

$$\sum \vec{F}_y = m\vec{a}_y = 0$$

y+

$$T cos\theta - mg = 0 => T = \frac{mg}{cos\theta} \ (4)$$



From eq(3) and (4):

$$=> F_{fictitious} = mg tan\theta \ (**)$$

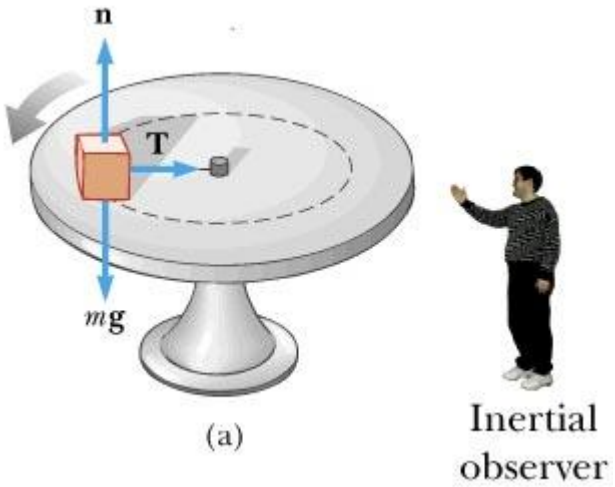
From eq(*) and (**):

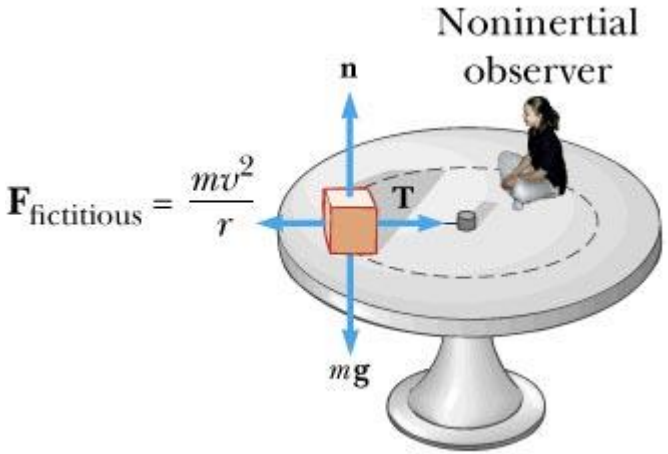
$$\Rightarrow F_{fictitious} = mgtan\theta = ma$$

$$\Rightarrow \ F_{fictitious} = ma$$

Fictitious force

- fictitious forces act in the direction opposite that of the acceleration of the noninertial frame





For inertial observer: the block moving in a circle. The stretched spring causes a **centripetal force**

For inertial observer: the block is at rest

⇒ fictitious force acting outward to balance with the tension.

⇒ This fictitious force is in opposite direction with the centripetal acceleration

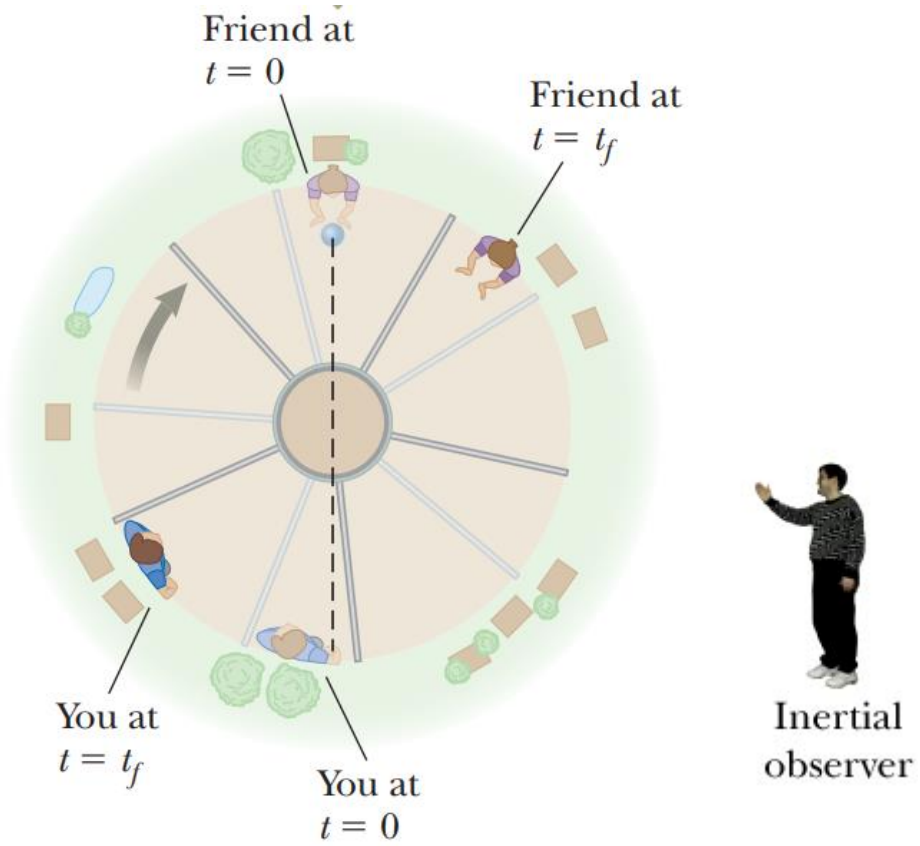
https://www.ux1.eiu.edu/~addavis/1350/06CirMtn/AccFrames.html

Fictitious force –Coriolis force

Suppose you and a friend are on opposite sides of a rotating circular platform and you decide to throw a baseball to your friend.

From inertial observer: the ball is in straight path

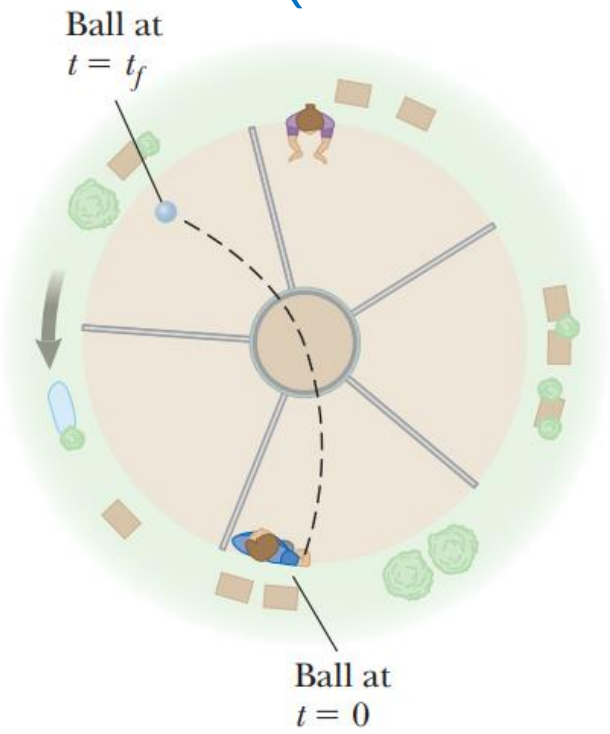
When the ball arrives at the other side, your friend is no longer there to catch it.



Your friend's point of view (non-inertial observer):

the ball veers to one side during its flight.

=> This deviation from the straight-path is due to fictitious force (Coriolis force)



Fictitious force-Coriolis force

https://www.youtube.com/watch?v=mcPs_OdQOYU&ab_channel=ScienceClicEnglish

- An observer in a noninertial frame of reference must account for fictitious forces to apply Newton's laws.
- When these fictitious forces are accurately defined, the motion described from a noninertial frame is equivalent with that described by an observer in an inertial frame.

Motion in the Presence of Resistive Forces

- When the object moves in the gas or liquid, the medium exerts a resistive force $\vec{R}$ on the object moving through it.
- The magnitude of $\vec{R}$ depends on the speed of the object
- The direction of $\vec{R}$ is always opposite the direction of the object's motion relative to the medium

Resistive Force Proportional to Object Velocity

The resistive force can be expressed as

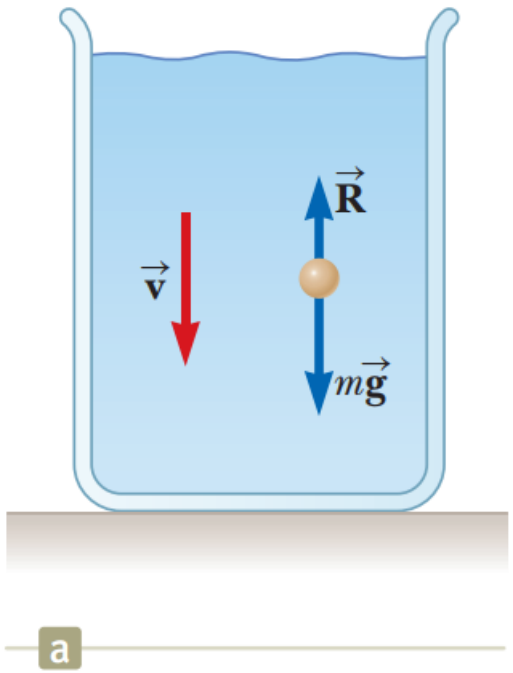
$$\vec{\mathbf{R}} = -b\vec{\mathbf{v}}$$

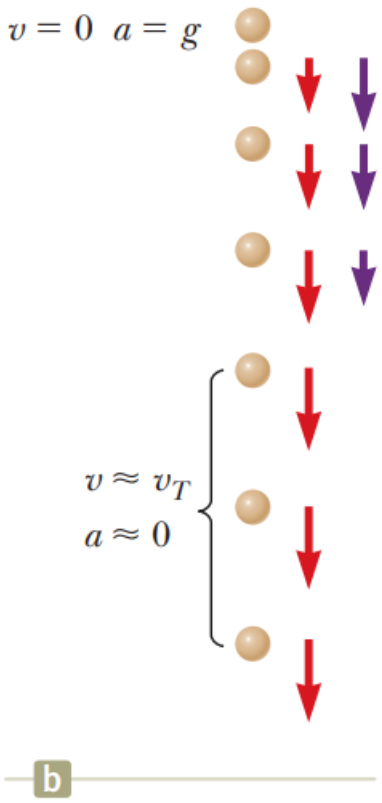
- where b is a constant, b depends on the properties of the medium and on the shape and dimensions of the object
- $\vec{v}$ is the velocity of the object relative to the medium
- Negative sign indicates that $\vec{R}$ is in the opposite direction to $\vec{v}$

Resistive Force Proportional to Object Velocity

Consider a small sphere of mass m released from rest in a liquid.

Assuming the only forces acting on the sphere are the resistive force $\vec{R} = -b\vec{v}$ and gravitational force $\overrightarrow{F_g}$





Resistive Force Proportional to Object Velocity

Applying Newton's second law to the vertical motion of the sphere:

$$\sum \overrightarrow{F_y} = m\vec{a}$$

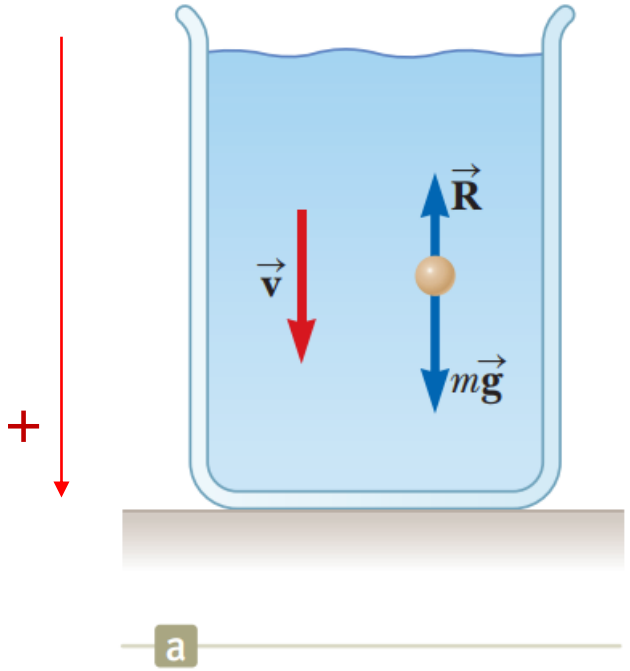
$$\Leftrightarrow \vec{F}_g + \vec{R} = m\vec{a}$$

$$+ \downarrow \quad F_g - \mathrm{R} = ma$$

Where R = bv and F_g = mg

$$\Rightarrow a = \frac{\mathrm{mg} - \mathrm{bv}}{m}$$

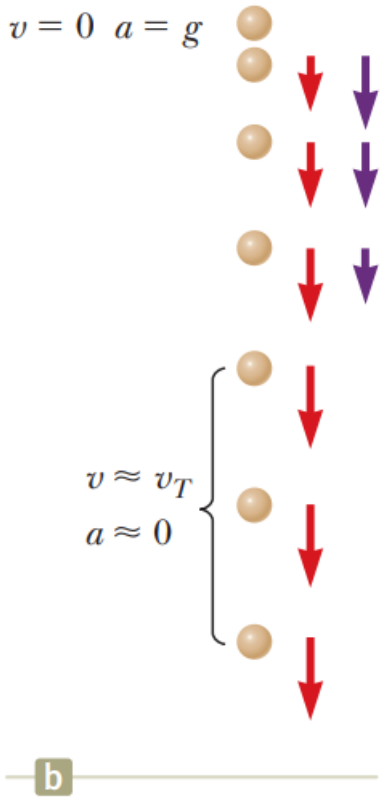
$$\Leftrightarrow \frac{dv}{dt} = \frac{\mathrm{mg} - \mathrm{bv}}{m}$$



Resistive Force Proportional to Object Velocity

- As t increases, the magnitude of the resistive force increases and the acceleration decreases.
- The acceleration approaches zero when the R approaches the F_g so that the net force on the sphere is zero and the speed of the sphere approaches its terminal speed v_T

$$a = g - \frac{bv_T}{m} = 0 \Rightarrow \boxed{v_T = \frac{mg}{b}}$$

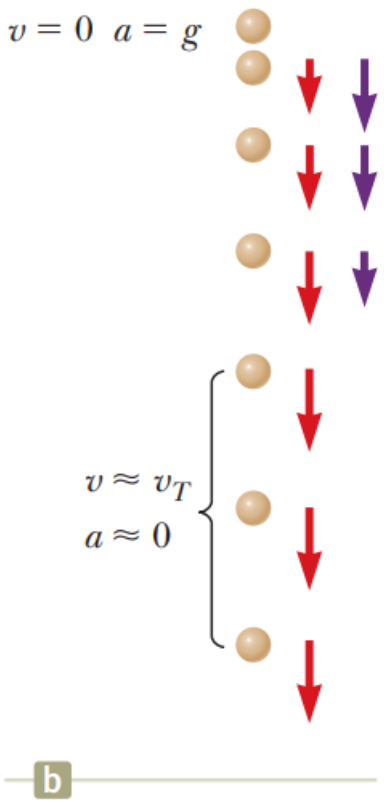


Resistive Force Proportional to Object Velocity

The expression for v for all times t:

$$\frac{dv}{dt} = \frac{mg - bv}{m} \Rightarrow \frac{dv}{\frac{mg - bv}{m}} = dt$$

$$\Leftrightarrow \frac{m}{mg-bv} dv = dt$$



Integrating both sides of the eq above:

$$\int_0^v \frac{m}{mg-bv} dv = \int_0^t dt$$

$$\Leftrightarrow \frac{-m}{b} ln(mg - bv)\Big|_0^v = t \Leftrightarrow \frac{-m}{b}[\ln(mg - bv) - \ln(mg)] = t$$

$$\Leftrightarrow \ln\left(\frac{mg-bv}{mg}\right) = \frac{-bt}{m} \Leftrightarrow \frac{mg-bv}{mg} = e^{\frac{-bt}{m}} \Leftrightarrow mg - bv = mge^{\frac{-bt}{m}}$$

Resistive Force Proportional to Object Velocity

$$mg - bv = mge^{\frac{-bt}{m}}$$

$$\Leftrightarrow v = \frac{mg}{b}\left(1 - e^{\frac{-bt}{m}}\right) \qquad where\ v_T = \frac{mg}{b}$$

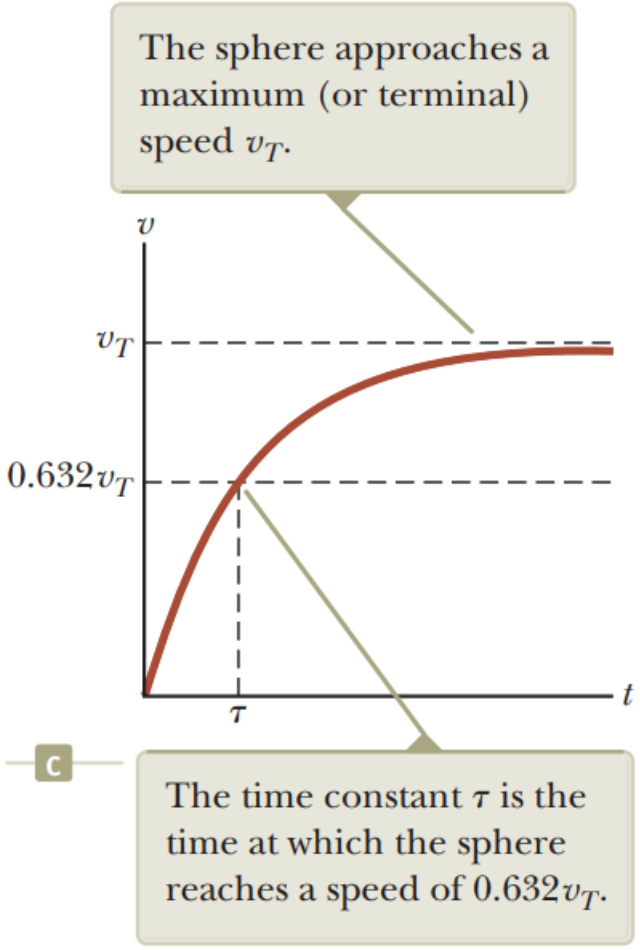
=> $$\boxed{v = v_T\left(1 - e^{\frac{-t}{\tau}}\right)}$$ Where $\tau = \frac{m}{b}$

$\Rightarrow$At t= 0: v =0

$\Rightarrow$At t = τ: v= 0.632v_T

$\Rightarrow$At t → ∞ : v=v_T

The v_T is proportional to the mass of the object



Resistive Force Proportional to Object Speed Squared

- For objects moving at high speeds through air, such as airplanes, skydivers, cars, and baseballs

=> the magnitude of the resistive force can be expressed as:

$$R = \frac{1}{2} D \rho A v^2$$

where *D: drag coefficient,*
r: density of air
A: cross-sectional area of the moving object measured in a plane perpendicular to its velocity.
v: the speed of the object

Resistive Force Proportional to Object Speed Squared

Suppose an object of mass *m* is released from rest:

$\Rightarrow$ Apply Newton's second law:

$$\sum \vec{F} = m\vec{a}$$

$$\Leftrightarrow \vec{R} + \vec{F_g} = m\,\vec{a}$$

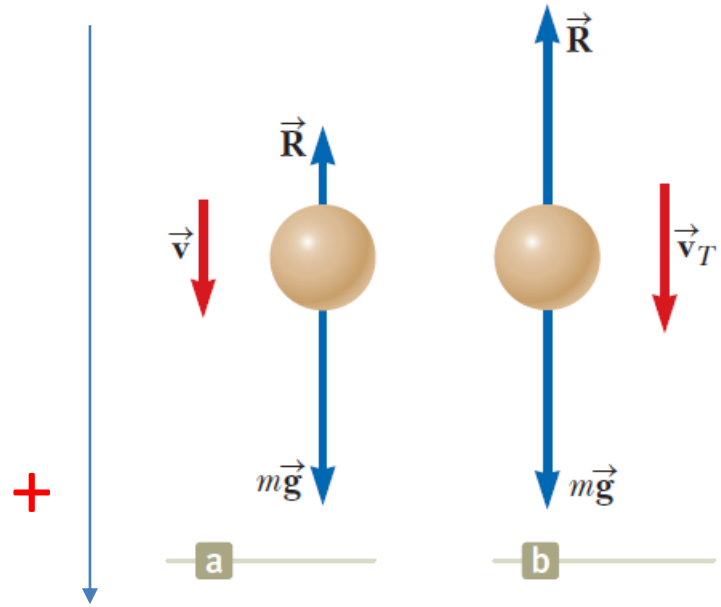
$+\downarrow$: mg-R = ma $\Leftrightarrow$ mg-$\frac{1}{2}D\rho A v^2 = ma$

$\Leftrightarrow$ a = g - $\frac{1}{2m}D\rho A v^2$

When a = 0, $\sum \vec{F} = 0, v = v_T$:

=> g - $\frac{1}{2m}D\rho A {v_T}^2 = 0$ $\Leftrightarrow$ $\boxed{v_T = \sqrt{\frac{2mg}{D\rho A}}}$

$$R = \frac{1}{2}D\rho A v^2$$



Resistive Force Proportional to Object Speed Squared

$$R = \frac{1}{2} D \rho A v^2$$

⇒The faster the speed of an object and larger the cross-sectional area the larger the resistive force

⇒ The slower the speed of an object and smaller the cross-sectional area the smaller the resistive force

$$v_T = \sqrt{\frac{2mg}{D\rho A}}$$

⇒The heavier the object and the smaller the cross-sectional area the larger the terminal speed

⇒The lighter the object and bigger the cross-sectional area the smaller the terminal speed

Motion in the Presence of Resistive Forces

Resistive Forces $\vec{R}$

The magnitude → depends on the medium

The direction → opposite the direction of motion of the object relative to the medium

Increases with increasing speed

Slow-speed objects

$$\vec{R} = -b\vec{v}$$

b depends on the property of the medium, and on the shape and dimensions of the object

high-speed objects

$$R = \frac{1}{2} D\rho A v^2$$

D → *drag coefficient*

ρ → the density of air

A → the cross-sectional area of the object

v → the speed of the object

HOMEWORK

6. A car initially traveling eastward turns north by traveling in a circular path at uniform speed as shown in Figure P6.6. The length of the arc ABC is 235 m, and the car completes the turn in 36.0 s. (a) What is the acceleration when the car is at B located at an angle of 35.0°? Express your answer in terms of the unit vectors $\hat{\mathbf{i}}$ and $\hat{\mathbf{j}}$. Determine (b) the car's average speed and (c) its average acceleration during the 36.0-s interval.

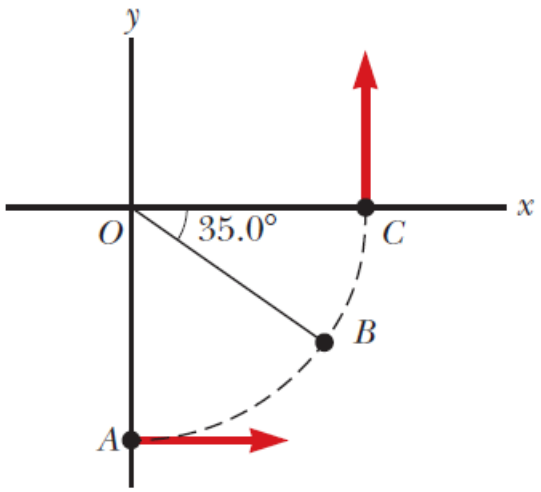


Figure P6.6

8. Consider a conical pendulum (Fig. P6.8) with a bob of mass $m = 80.0$ kg on a string of length $L = 10.0$ m that makes an angle of $\theta = 5.00°$ with the vertical. Determine (a) the horizontal and vertical components of the force exerted by the string on the pendulum and (b) the radial acceleration of the bob.

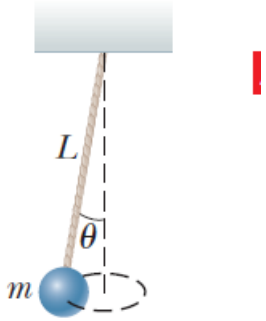


Figure P6.8

HOMEWORK

18. One end of a cord is fixed and a small 0.500-kg object is attached to the other end, where it swings in a section of a vertical circle of radius 2.00 m as shown in Figure P6.18. When $\theta = 20.0°$, the speed of the object is 8.00 m/s. At this instant, find (a) the tension in the string, (b) the tangential and radial components of acceleration, and (c) the total acceleration. (d) Is your answer changed if the object is swinging down toward its lowest point instead of swinging up? (e) Explain your answer to part (d).

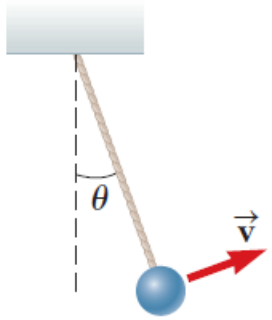


Figure P6.18

21. M An object of mass $m = 0.500$ kg is suspended from the ceiling of an accelerating truck as shown in Figure P6.21. Taking $a = 3.00$ m/s^2, find (a) the angle θ that the string makes with the vertical and (b) the tension T in the string.

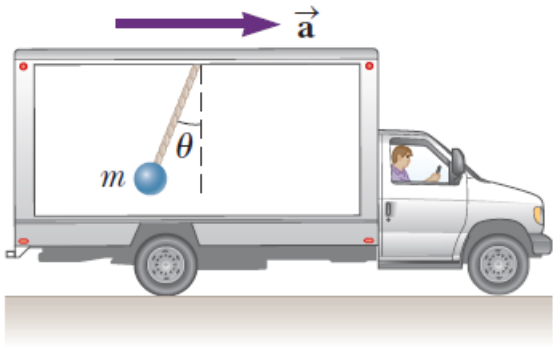


Figure P6.21

23. M A person stands on a scale in an elevator. As the elevator starts, the scale has a constant reading of 591 N. As the elevator later stops, the scale reading is 391 N. Assuming the magnitude of the acceleration is the same during starting and stopping, determine (a) the weight of the person, (b) the person's mass, and (c) the acceleration of the elevator.

HOMEWORK

26. Review. (a) Estimate the terminal speed of a wooden sphere (density 0.830 g/cm^3) falling through air, taking its radius as 8.00 cm and its drag coefficient as 0.500. (b) From what height would a freely falling object reach this speed in the absence of air resistance?

28. A skydiver of mass 80.0 kg jumps from a slow-moving aircraft and reaches a terminal speed of 50.0 m/s. (a) What is her acceleration when her speed is 30.0 m/s? What is the drag force on the skydiver when her speed is (b) 50.0 m/s and (c) 30.0 m/s?

The end ☺